Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

KitaHack

Selected Domain

Domain 1: The Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 4: Quality Education (target 4.4)

Project / Solution Name

FuturePath

Name (Leader)

CHANTHIRAA CHELVAN PASUPATHY

Email & Phone (Leader)

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University Affiliation(s)

Universiti Teknikal Malaysia Melaka (UTeM)

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian students, fresh graduates and early-career workers must choose careers in a job market that AI and automation are reshaping faster than career advice can keep up. Employers surveyed for the WEF Future of Jobs Report 2025 expect 39% of workers' existing skills to be transformed or outdated by 2030, and estimate that 59 in every 100 workers will need training. In Malaysia, a 2024 Human Resources Ministry and TalentCorp study found that about 600,000 workers in 10 strategic sectors need reskilling within three to five years. Clerical jobs such as data entry clerks, bank tellers and administrative assistants are among the fastest-declining roles, while AI, data and green-energy roles grow.

The people most affected have the least access to personal guidance: students deciding what to study, graduates whose CVs list courses rather than skills, and workers in at-risk roles who do not know which of their skills transfer. Job portals match on job titles, so people cannot see which skills they already have, how far they are from a target role, or which affordable course would close the gap. Generic advice also ignores how fast AI adoption may move.

The cost is social and economic: time and money spent on the wrong training, skills mismatch for employers, and a risk that the AI transition widens inequality instead of creating decent work. People need transparent, evidence-based guidance they can act on.

Sources: WEF Future of Jobs Report 2025 (weforum.org); Bernama, 10 October 2024.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FuturePath is a web app that turns a CV into a personalised, explainable career and reskilling plan for Malaysian students and workers.

1. Read the CV with AI. Users upload a CV (PDF, DOCX or TXT) or build a profile by hand. Qwen2.5-3B-Instruct, running on a RunPod Serverless GPU, extracts skills from a 110-skill catalogue, and every suggestion must quote the CV exactly; anything it cannot quote is rejected. A keyword dictionary fills gaps if the GPU is unavailable, and Claude reads image-only PDFs. Users confirm each skill and its level before anything is scored.

2. Rank careers with live evidence. A FastAPI engine compares confirmed skills with 44 careers and combines skill match with market demand built from dated trend signals: 36 curated reports (WEF, ILO, Malaysian sources) plus news and research feeds that a daily job checks and verifies. An optional RIASEC interest quiz adds interest fit. Users can switch between Moderate and Fast AI-disruption scenarios; in the Fast scenario, AI-related signals count double, so rankings visibly change. Every score shows its formula and sources.

3. Plan the learning. For a chosen career, FuturePath lists missing and below-level skills and builds a week-by-week pathway from 111 free, low-cost and HRD Corp-claimable courses, linked to the relevant SDGs.

FuturePath applies the SDC Phase 1 workshop skills directly: **GenAI APIs** (RunPod-hosted Qwen and the Claude API, with every model reply validated against the CV), **database architecture** (SQLite for profiles, signals and review history, plus versioned JSON catalogues) and **cloud hosting** (Replit Autoscale for the web app, RunPod Serverless for GPU inference and GitHub Actions for the daily evidence check). The interface is available in English, Malay and Chinese.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: HTML, CSS and JavaScript with a React runtime, served by FastAPI; English, Malay and Chinese.

Backend: Python 3.12, FastAPI, Pydantic, Uvicorn and httpx; pypdf and python-docx read CVs.

AI: Qwen2.5-3B-Instruct on RunPod Serverless (CV skill extraction, JSON output checked against the CV text); Anthropic Claude API (image-only PDFs, verifying new trend signals); keyword dictionary as a fallback.

Database: SQLite for profiles, signals, the review queue and audit history; versioned JSON catalogues of 44 careers, 110 skills, 111 courses and 36 curated signals.

Cloud: Replit Autoscale (hosting and Secrets), RunPod Serverless (GPU), GitHub and GitHub Actions (source control, CI tests, daily verified-signals snapshot). Testing uses pytest and Playwright.

Flow: browser > FastAPI on Replit > RunPod Qwen or Claude > quote validation > user confirms skills > matching engine (SQLite and JSON data, daily signals from GitHub Actions) > career ranking, learning pathway and SDG links. *See Appendix A for the block diagram.*

Sponsor resources: Yes. RunPod: USD 25 of sponsor-provided GPU credits run the Serverless endpoint that hosts Qwen2.5-3B-Instruct for CV skill extraction. **Replit:** complimentary Replit Premium (SDC sponsorship) is used to develop, configure and host the FastAPI app at https://futurepathkitahack.replit.app.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we started from three users in the Domain 1 brief: a student choosing a field, a graduate with a course-heavy CV, and a worker in an at-risk clerical role. Each needed to see their own evidence, the reason behind every score, and an affordable next step.

Define: turn a CV into confirmed skills, then into a ranked, explained career and learning plan.

Ideate and prototype: we built in short loops and reviewed each working page as a team. Feedback shaped key decisions: AI-suggested skills stay unconfirmed until the user approves them, below-level skills show the user's level against the required level, each score shows its formula and sources, and long lists use a See more button instead of dense charts.

Test: more than 280 automated pytest checks and Chromium browser journeys cover CV upload, the quiz, rankings, pathways, mobile layout and the password-protected demo.

Deploy: the app already runs on Replit with RunPod GPU inference, secrets kept on the server, a keyword fallback when the GPU is cold and a daily evidence check, so Phase 2 can focus on measured user testing with students rather than rebuilding.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." FuturePath helps each user find which of their skills are relevant to growing careers and which courses close the gap. **Secondary:** Target 8.6, reduce the proportion of youth not in employment, education or training.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Global:** 39% of workers' existing skills are expected to be transformed or outdated by 2030, and 59 in 100 workers will need training (WEF Future of Jobs Report 2025). **Malaysia:** about 600,000 workers in 10 strategic sectors need reskilling within 3 to 5 years (Human Resources Ministry and TalentCorp study; Bernama, 10 October 2024). **Prototype baseline, measured before Phase 2 testing:** the precision of keyword-only skill matching on 20 labelled fictional CVs, and the median minutes a participant needs to build a skills-gap and course plan by hand.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	**1. Grounded extraction precision:** % of AI-suggested skills that are correct and backed by an exact CV quote, on 20 pre-labelled fictional CVs. **2. Time to plan:** median minutes from CV upload to a confirmed career shortlist and learning pathway, against the manual baseline. **3. Usefulness:** % of test participants who rate their pathway 4 or 5 out of 5.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**1.** At least 80% grounded skill precision, and at least 15 points above the keyword-only baseline. **2.** At least 30% less median time to produce a skills-gap and learning plan than the manual baseline. **3.** At least 70% of 20 or more student testers rate their pathway useful (4 or 5 out of 5). *Test sets are frozen before evaluation; results will report sample size and fallbacks.*

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. No code, design, research or other work created or owned by a team member before the hackathon is included. Third-party resources are declared in 6.2.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI, Uvicorn, Pydantic, httpx	Code: Python web framework, server, data validation and HTTP client	MIT (FastAPI, Pydantic); BSD-3-Clause (Uvicorn, httpx)
pypdf, python-docx, pytest, Playwright	Code: CV parsing (PDF and DOCX) and automated testing	BSD-3-Clause (pypdf); MIT (python-docx, pytest); Apache-2.0 (Playwright)
React, ReactDOM, Babel	Code: browser UI runtime	MIT
Qwen2.5-3B-Instruct (via RunPod); Anthropic Claude API	Model: pre-trained LLMs, used without fine-tuning, for CV skill extraction, image-only PDFs and signal verification	Qwen Research License; Anthropic Commercial Terms
WEF, ILO, Google News RSS, arXiv, course providers	Dataset: public reports, dated news and research feeds, course pages; curated and cited	Each source's own terms; cited with links

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. No team member is bound by an employment, internship or consulting agreement that could give an employer a claim to this work.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

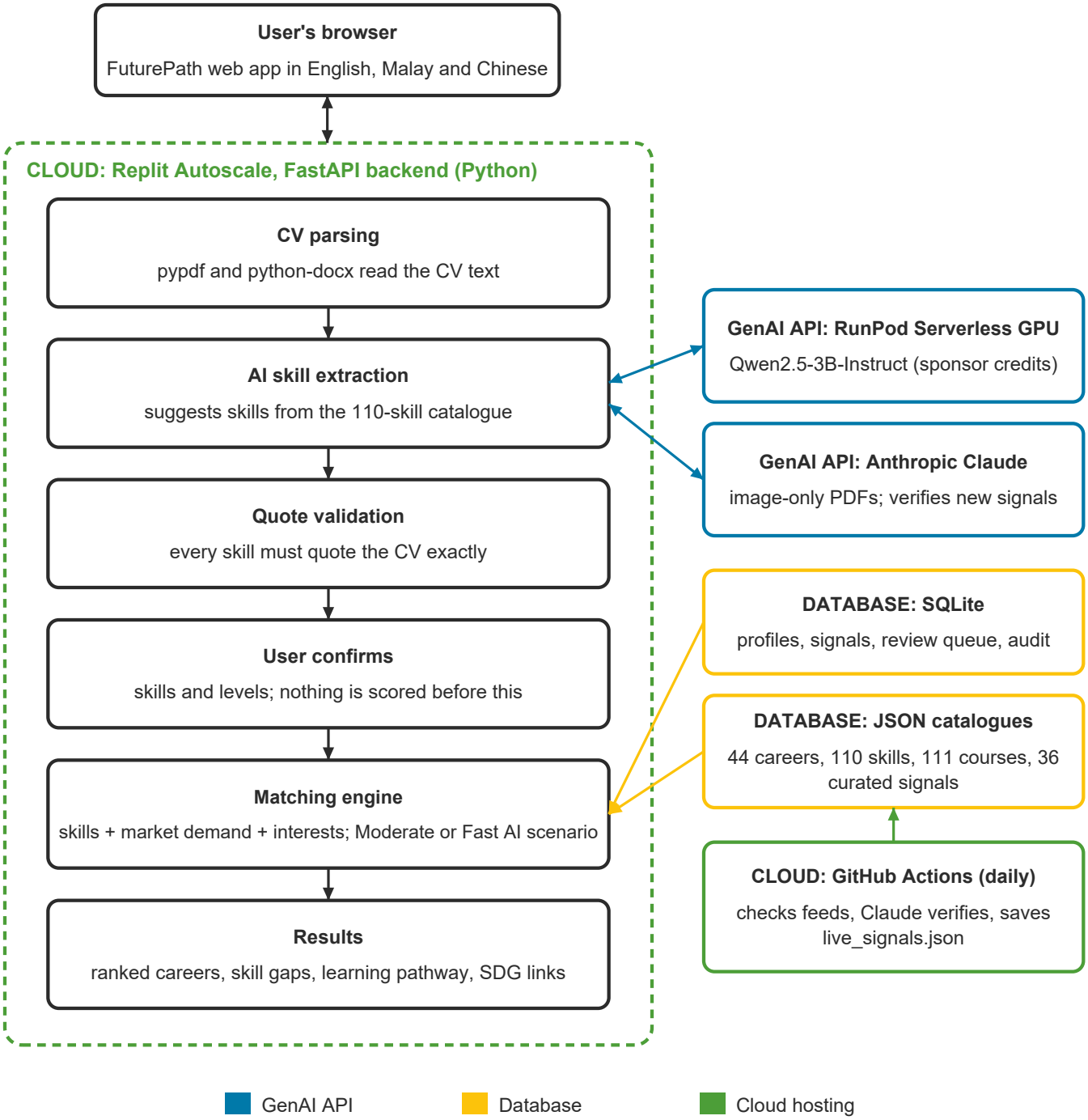


Date – 24 October 2026

APPENDIX A

SYSTEM BLOCK DIAGRAM

How the SDC Phase 1 workshop elements, **GenAI APIs**, **database** and **cloud hosting**, connect into one FuturePath architecture.



Arrows show data flow. Secrets stay on the server (Replit Secrets); the browser never calls RunPod or Claude directly. If the GPU is unavailable, a keyword dictionary replaces the AI step.